

Signal Processing

Theorems and Correspondences of the Fourier transform

Properties, Theorem	Time domain	Frequency domain
Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
Differentiation	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$
Integration	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
Time domain scaling	$x(at), a \neq 0$	$\frac{1}{ a } X\left(\frac{j\omega}{a}\right)$
Frequency domain scaling	$\frac{1}{ b } x\left(\frac{t}{b}\right)$	$X(bj\omega), b \neq 0$
Convolution	$x(t) * y(t)$	$X(j\omega) \cdot Y(j\omega)$
Multiplication	$x(t) \cdot y(t)$	$\frac{1}{2\pi} X(j\omega) * Y(j\omega)$
Time domain shift	$x(t + a)$	$e^{j\omega a} X(j\omega)$
Frequency domain shift	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$

Time domain	Frequency domain
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$\dot{\delta}(t)$	$j\omega$
$\sigma(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$\text{rect}(at)$	$\frac{1}{ a } \text{si}\left(\frac{\omega}{2a}\right)$
$\text{tri}(at)$	$\frac{1}{ a } \cdot \text{si}^2\left(\frac{\omega}{2a}\right)$
$\text{si}(at)$	$\frac{\pi}{ a } \text{rect}\left(\frac{\omega}{2a}\right)$
$\frac{1}{t}$	$-j\pi \cdot \text{sgn}(\omega)$
$\text{sgn}(t)$	$\frac{2}{j\omega}$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos(\omega_0 t)$	$\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$
$\sin(\omega_0 t)$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
$e^{-a t }, a > 0$	$\frac{2a}{a^2 + \omega^2}$